POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Object Oriented Programming in C++ (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why is OOP considered a new paradigm in programming language? Justify with OOP important features. 7

   b) What do you mean by programming in small and programming in large? Explain abstraction mechanism in C++ with suitable examples. 8

2. a) What is a constructor? Explain different types of constructors in C++ with suitable programs. 8

OR

List out the characteristics of static data member. Explain static member function with example.

   b) What is friend function? Write a program to Swap data between two classes using friend function. 7

3. a) Define dynamic memory allocation. How can dynamic memory allocation be achieved in C++? Explain. 7

   b) How does inheritance support reusability? Justify your answer with suitable examples. 8

4. a) Create a class Animal with data members name, age, and species. Create another class WildAnimal with data members region, diet, and conservation_status. Additionally, create another class PetAnimal with data members owner_name and favorite_food. The PetAnimal and WildAnimal classes are inherited from the Animal class. Each class should have at least one constructor that uses the base class constructor. Implement a showdata() member function in each class to display the respective information of the class members. 7

   b) What is operator overloading? Overload + operator to add two complex number. 8

5. a) What is type conversion? Write a program to convert the object of polar class to rectangular class with the help if type conversion. 8

OR

Differentiate between function overloading and function overriding with suitable examples.

b) What do you mean by generic programming? Create a function template to perform addition of two integers and two float. 7

6. a) What is an exception? How is it handled in C++? Write a program to handle exceptions with multiple catch blocks. 7

b) Write a C++ program to input record of 'n' students (name, address and roll) and save to a file. Display the record of only those student whose address is "Kathmandu" after reading from file. 8

7. Write short notes on: **(Any two)** 2×5

a) Composition

b) Subtype vs subclass

c) STL